USER GUIDE / PATCH 64

Targeting Pod for UH-60L DAP

AN/AAQ-16D-inspired TV / IR camera and coded laser designator

English operating guide | Revision 1.1 | 28 October 2026
Applies to submod patch 64, including its inherited features.

Simulation and AI disclosure

This is an unofficial gameplay modification. All parameters, interface elements, operating modes and functions are highly approximate. Many values and design choices are invented for gameplay convenience rather than derived from verified technical data. The real AN/AAQ-16D is an inspiration, not an accurately reproduced system.

Generative AI was used in creating the mod and this documentation, including assistance with code, assets, research and writing. AI-assisted output can contain errors. This guide is for this game modification only and is not a real-world equipment or weapons operating manual.

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01 / SETUP

Installation and first use

Install the complete package

Close DCS completely. Extract UH60_DAP_AN_AAQ16D_patch64_FULL.zip into the UH-60L mod root and allow file replacement. The archive contains the complete game overlay and this PDF in Documentation; earlier patches are not required. The target folder must contain the mod entry.lua and Cockpit folder, not an extra nested copy of the mod.

This submod targets UH-60L DAP. Select the DAP aircraft and its DAP control profile. The ordinary UH-60L is not the target. Compatibility with other overlays such as Armed Blackhawk has not been established; use a clean base mod for a reproducible installation.

Fit the pod and configure the aircraft

In the Mission Editor / loadout, fit AN/AAQ-16D to station 8. The code checks both the station identity and a positive pod inventory count. Set TV/FLIR Laser code in aircraft additional properties if a code other than 1688 is required. No in-flight code-entry control is implemented.

Starting condition	What to expect
Hot / airborne start with pod	Ready in TV, BORE and WFOV. Laser off. IR preferences start at WHOT and MAN; image controls start at 50%.
Cold start with pod	POWER OFF. Press AN/AAQ-16D Power to begin a 60-second BIT, then enter TV / BORE / WFOV.
No pod or wrong pod	Black screen without operating labels. Pod commands are inactive; stock weapons remain independent.
Pod removed / reinstalled	Removal resets laser, POINT and countdowns. Reinstallation starts a new 60-second BIT.
Power cycled	Switching on from OFF runs BIT again. Camera pose and zoom reset; do not rely on retention of a target.

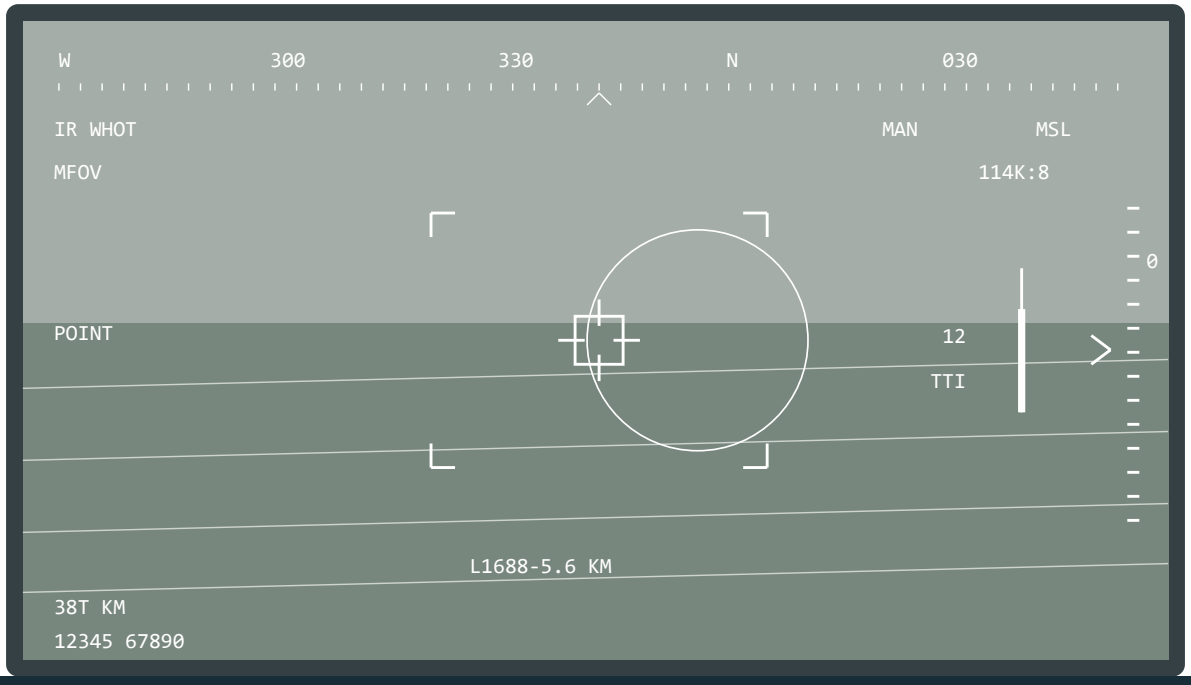
Minimum first-use sequence

1. Bind the pod controls in DCS Options > Controls > UH-60L DAP, category AN/AAQ-16D.
2. With the pod ready, select STAB and slew using buttons or a centered mini-stick.
3. Step through WFOV / MFOV / NFOV to identify the area of interest.
4. Toggle POINT for a ground-point hold; confirm the square around the reticle.
5. Select TV or IR, adjust the image, and use the laser as required.
6. For weapon release use the normal DAP weapon controls; pod cues are advisory.

The startup mask remains black until the camera has produced a valid draw. Window Show/Hide only hides the display; it does not shut down the laser.

02 / DISPLAY

Display reference



Element	Meaning
TV / IR WHOT / IR BHOT	Selected sensor channel and IR polarity. WFOV / MFOV / NFOV below it identifies the selected optical field.
AUTO / MAN	IR exposure state, shown only in IR. MAN is the fixed-exposure option.
MSL / 114K:8 / APKWS:8	Stock weapon selector in MSL; type of the next eligible missile and total remaining missiles of that family.
BORE / STAB / POINT	Current camera operating mode. A square bisecting the four reticle strokes means an active POINT hold.
Central cross and four corners	Cross marks the optical axis. Corner brackets show the next narrower FOV; absent at NFOV and while zoom is transitioning.
Top heading tape	Magnetic-heading reference from helicopter navigation. Cardinal labels use letters. Center marker references aircraft heading; the chevron shows pod direction.
Right elevation ladder	Static -55 to +20 degree scale, ticks every 5 degrees; longer 10-degree ticks and an emphasized 0. Moving right-pointing chevron shows pod elevation.
L1688-5.6 KM / MGRS	Live laser code and rounded slant range; held ground-point grid at lower left. See page 6.
Circle / TTI	Hellfire-only advisory launch geometry and estimated time to impact. See page 7.

Symbols use white strokes with a black outline for scene contrast. Illustration is schematic; symbol placement and scale may differ slightly from the live output.

03 / CAMERA CONTROL

Modes, steering and Zeroing

Mode	Operation
BORE	Camera returns to body-forward azimuth 0 / elevation 0 and stays fixed relative to the helicopter. Slew is locked. Zoom and sensor selection remain available.
STAB	Maintains an inertial viewing direction while the helicopter rotates. Slew demand is applied to this stabilized direction, reducing the need to fight helicopter motion. Translation changes the viewed ground point: STAB is not a fixed ground lock.
POINT	Holds one world-space terrain point as the helicopter translates and rotates, within mechanical limits. The active-lock square is shown. This is not image correlation or moving-object tracking.

Mode selection and temporary slewing

Mode BORE STAB alternates BORE and STAB; pressing it from POINT returns to BORE. POINT Toggle arms or cancels ground-point hold. From BORE it first enters STAB. Acquisition occurs after controls settle at neutral for approximately 0.06 seconds.

Slewing during POINT temporarily releases the lock into STAB but keeps POINT intent armed. When the mini-stick/buttons return to neutral, a new ground intersection is acquired. A failed acquisition leaves STAB and cancels that intent. At a mechanical POINT limit, the system also changes to STAB, cancels reacquisition and flashes the diagonal X twice over one second.

Cage and mechanical limits

Cage cancels POINT intent, commands body-forward 0/0 and resets WFOV. In BORE it stays BORE; in STAB it remains STAB. From POINT it releases into STAB. Cage is not the same as a permanent BORE lock.

Commanded travel is azimuth -210 to +210 degrees and elevation -55 to +20 degrees, relative to the aircraft. These are implementation limits, not certified specifications. The external cup and sensor head follow the applied camera angles. Aircraft structure may still obstruct a view; no complete physical masking model is promised.

Zeroing: cue from the navigation system

Zeroing reads the current Fly-To waypoint of the powered ASN-128, from any operating mode. It is waypoint cueing, not boresight calibration. If the direction is reachable, the camera slews there and tries to hold the first terrain intersection along that ray in POINT. This intersection need not equal the stored waypoint altitude or exact waypoint position.

If the ray has no valid terrain point, the camera remains in STAB. An invalid/unavailable waypoint or unreachable direction produces a large diagonal X at 2 Hz for one second. No navigation database is copied into the pod; the stock ASN remains authoritative.

STAB does not remove image roll. POINT queries terrain rather than visible object surfaces; see page 6.

04 / IMAGE

TV / IR and image adjustment

Channels and fields of view

TV / IR Channel alternates a monochrome daylight TV view and the DCS-derived IR rendering path. TV is not a light-amplified night-vision channel. IR appearance comes from the game renderer; it is not a calibrated thermal camera or radiometric measurement.

Field	Horizontal FOV	Use in this mod
WFOV	42 degrees	Search and orientation
MFOV	6.7 degrees	Identification and aiming
NFOV	2.5 degrees	Narrow view / fine pointing

TV and IR currently share the same three FOV steps. Each Zoom In / Out press selects one step; holding does not repeatedly cycle. Transition time is 0.5 seconds. The four-corner preview uses perspective FOV ratios, not a fixed magnification label. There is no separate continuous digital zoom control.

Steering speed follows zoom

Full-rate slew is approximately 140 degrees/second at WFOV, scaled by current FOV / 42. MFOV is about 22.3 degrees/second and NFOV about 8.3 degrees/second. Mini-stick deflection scales speed further; returning to center stops manual demand. These are commanded maxima, not measured turret performance.

Two separate layers of image controls

Control	Range and behavior
Brightness / Contrast	Final screen correction for the camera image. Default 50%, limited to 20-80%. Separate increase/decrease commands, a combined reset, and absolute axes.
IR Gain / IR Level	Independent native IR processing parameters. Default 50%, limited to 20-80%, with individual resets and absolute axes. Gain changes IR tonal separation; Level shifts its operating level. Percentages are normalized controls, not detector units.
WHOT / BHOT	IR polarity: hotter-looking output is light in WHOT and dark in BHOT. Selects native rendering passes; it is not a CPU pixel-inversion filter.
AUTO / MAN	IR exposure toggle. AUTO permits native scene-dependent exposure. MAN uses a fixed exposure setting to reduce exposure pumping as sky or aircraft structure enters/leaves the frame. It does not keep average screen brightness constant.

Brightness/contrast and gain/level buttons change by 2 percentage points on press. After a 0.35-second hold they repeat, accelerating from about 4 to 24 percentage points/second over two seconds. Gain/level readouts appear for about three seconds after adjustment, only in IR. Absolute axes map -1 / 0 / +1 to 20 / 50 / 80%.

Fixed exposure is the initial IR preference. It cannot restore detail already clipped by the renderer. Surface appearance, weather, scene composition and DCS changes can still affect the image.

05 / DESIGNATION

Laser, range and coordinates

Coded laser

The coded laser originates at the pod mounting position and follows the current optical axis in BORE, STAB and POINT. The default code is 1688. Set a different code before the mission using the aircraft property TV/FLIR Laser code. Invalid or missing code values fall back to 1688. The implemented valid code pattern is 1[1-7][1-8][1-8].

Two independent inputs request emission:
Laser Toggle latches on/off.
Fire Weapon First Detent, the existing DAP action, requests laser while held.
They are combined with OR logic: releasing the first detent does not extinguish a laser that is still toggled on. To stop emission, clear the toggle and release the detent. The stock detent command is still forwarded normally.

What the laser indication confirms

The L-code appears only when the native spot reports active emission at a valid terrain intersection. A toggle request alone does not guarantee a visible L-code. No terrain intersection, sky pointing, power loss or pod removal suppresses the spot and its indication. A maintained request may recover automatically when a valid intersection returns.

Readout	Interpretation
L1688-5.6 KM	Code 1688; estimated slant distance from the pod to the designated terrain point, rounded to 0.1 km.
L1688-XXX	The spot is active but range is unavailable or exceeds the 16 km display limit. This is not a 16 km weapon range claim.
No L-code	No acknowledged active spot. Check power, pod presence, laser request and terrain intersection.

POINT coordinates

In active POINT, the lower-left field shows the held terrain point in MGRS, for example 38T KM 12345 67890, split over two lines. MGRS is available independently of weapon selection. It is cleared when POINT is released. A failed map conversion displays a placeholder rather than invented coordinates.

Displayed precision does not establish positional accuracy. Terrain sampling, camera origin and the simulated navigation/rendering environment limit the result. Range and coordinates are calculated by the mod, not read from a simulated real laser rangefinder.

Terrain-only designation

Acquisition marches along the ray in 25 m steps out to 20 km and refines a detected terrain crossing. It does not trace vehicle bodies or building roofs. A visible object in the crosshair can therefore differ from the actual held/designated terrain point. Narrow terrain features can also be missed.

06 / WEAPONS

Missile cues and countdown

MSL and ammunition

MSL mirrors the stock weapon selector. Below it, 114K:N or APKWS:N identifies the next eligible missile family and the total remaining count of that family across all four weapon stations, including electrically unselected stations. It is not the count on just the next rack. The display updates after release and rearming. Unknown/other missiles do not receive one of these labels.

The observer follows the stock station priority (internal station indices 0, 6, 1, 5). The pod does not choose a rack, command missile seeker pointing, set a launch trajectory, or provide seeker-lock acknowledgement. 114K is the display label used for the identified Hellfire family; it is not variant verification.

Advisory launch circle

With MSL selected and the next eligible missile positively identified as Hellfire, the pod screen displays a projected 7.5-degree half-angle cone about aircraft-forward. Its projected size changes with FOV, not target distance. A small central cross helps alignment. The circle is solid inside the angular criterion and dashed outside; edge cues indicate an offscreen reference.

In POINT, a related green circle and center cross are drawn on the helicopter collimator when the held target is within the displayable forward sector. If the laser is active and valid unrounded range is 2.0 km or less, both circles are forced dashed and blink at 2 Hz. This is an approximate minimum-range warning.

No launch inhibition and no seeker guarantee

Since patch 62, the pod imposes no launch blocks. A warning does not prevent firing. The circle is a simple fixed-rail geometric aid: it does not predict seeker acquisition, blind-flight distance, loft, flight-path curvature or actual hit probability. The real DCS missile may miss inside the circle or hit outside it.

Estimated time to impact (TTI)

A confirmed stock Hellfire release, observed through inventory reduction, may start a countdown using range at release. Up to two most recent confirmed launches are retained. The display shows whichever is predicted to arrive first, then the other. A third launch discards the oldest launch entry. Only one countdown is displayed at a time.

Seconds appear right of the reticle, left of the elevation scale. A wide vertical fill overlays a thin full-height guide, about 20% of video height. The bottom edge stays fixed while the top edge descends as time runs out, shortening the fill from the top. At expiry the fill disappears; there is no short traveling marker. This is an empirical range-based estimate, not missile telemetry or impact detection. The model supports positive ranges through 9 km; a release with no usable range may produce no countdown.

TTI and circles hide when the next missile is APKWS/unknown or MSL is deselected. Hidden countdowns keep running and may return if the display conditions return. After the last Hellfire is fired, there may be no next Hellfire, so its countdown can disappear. Power-off or pod removal clears the stack. Missile destruction, a miss or an early impact is not detected.

07 / REFERENCE

Button commands

Find these actions in the AN/AAQ-16D category of the UH-60L DAP input profile. Bind your own keys/HOTAS buttons; do not assume universal defaults. The common prefix AN/AAQ-16D is omitted below.

Action name suffix	Function
Power	Toggle OFF / startup BIT. A new startup requires 60 seconds.
Window Show/Hide	Show or hide the camera overlay. Does not switch the pod or laser off.
Slew Left / Right / Up / Down	Hold for camera movement in STAB or temporary manual slew from POINT. BORE ignores slew.
Zoom In / Zoom Out	One WFOV / MFOV / NFOV step per press.
Cage	Cancel POINT, return to 0/0 and WFOV; retain BORE or STAB as described on page 4.
Mode BORE STAB	Toggle body-fixed BORE / direction-stabilized STAB.
POINT Toggle	Arm ground-point hold or cancel it into STAB.
Zeroing	Cue to current ASN-128 Fly-To direction; attempt POINT, otherwise STAB.
TV / IR Channel	Toggle monochrome TV / native IR path.
Laser Toggle	Latch laser request. First detent is a separate hold request.
Brightness Increase / Decrease	Adjust final screen brightness; hold accelerates.
Contrast Increase / Decrease	Adjust final screen contrast; hold accelerates.
Brightness and Contrast Reset	Restore both final-image controls to 50%.
IR Exposure AUTO / MAN	Select native automatic / fixed exposure.
IR Polarity WHOT / BHOT	Toggle IR output polarity.
IR Gain Increase / Decrease	Adjust normalized IR gain; hold accelerates.
IR Level Increase / Decrease	Adjust normalized IR level; hold accelerates.
IR Gain Reset / IR Level Reset	Reset the respective IR control to 50%.
Fire Weapon First Detent (stock DAP)	Hold to request laser in addition to its normal stock handling. This action is in the helicopter controls, not necessarily the pod category.

Aircraft weapon selection, arming and firing remain stock DAP functions. This guide does not replace the UH-60L base-mod manual.

08 / REFERENCE

Axes and practical bindings

Axis name suffix	Input model
Camera Azimuth Axis	Centered rate command. Left/right deflection changes horizontal slew speed; it is not an absolute turret angle.
Camera Elevation Axis	Centered rate command. Up/down deflection changes elevation slew speed.
Brightness Absolute Axis	Absolute 20-80% final-image brightness. Center = 50%.
Contrast Absolute Axis	Absolute 20-80% final-image contrast. Center = 50%.
IR Gain Absolute Axis	Absolute normalized IR gain, 20-80%.
IR Level Absolute Axis	Absolute normalized IR level, 20-80%.

Mini-stick setup

Assign the two camera axes to a self-centering mini-stick. The mod applies a 5% center dead zone and rescales the remaining travel. Greater deflection gives faster movement up to the current zoom-dependent maximum. Use DCS axis inversion if your hardware direction is reversed. Keep the stick centered when entering POINT.

Button slew and axis demand are combined and limited to full-rate demand. Avoid duplicate bindings on another connected controller: a stray axis outside the dead zone can release POINT or keep slewing. Add a DCS dead zone only as needed; excessive combined dead zones make small corrections harder.

Image sliders and knobs

Use sliders or rotary axes for absolute brightness, contrast, gain or level. The mod applies them on input events. A button/reset may override a stationary axis until that axis sends another event; noisy hardware can continually overwrite the button setting. Leave an absolute axis unbound if you prefer buttons.

Suggested compact HOTAS set

Assign a mini-stick for slew, two buttons or a rocker for zoom, and separate buttons for POINT, BORE/STAB and Cage. Add Zeroing, TV/IR and Laser Toggle where convenient. Keep first-detent laser and the normal weapon-release action clearly distinguished. Image controls can be placed on keyboard keys or spare knobs.

Useful control checks

With the helicopter stationary: select STAB, verify both axis directions, release to neutral, acquire POINT and confirm the square. Move the mini-stick, release it and check that POINT reacquires a new terrain location. Select BORE and check that slew no longer moves the camera. Use Cage to restore WFOV.

No separate slew-speed switch

The current implementation changes pointing rate through FOV and mini-stick deflection. There is no additional user-selectable slow/fast rate command in the patch 64 binding list.

09 / BOUNDARIES

Limitations and troubleshooting

Symptom / boundary	Check or explanation
Black screen without labels	Verify DAP, AN/AAQ-16D on station 8, positive pod count and a supported DCS build. Missing/unsupported native integration can also leave the fail-closed black state.
POWER OFF / BIT	Turn the pod on and allow the full 60-second startup. Rearming with the pod also starts BIT.
Camera will not slew	BORE deliberately locks steering. Select STAB, check DAP bindings, power and mini-stick assignment.
POINT will not acquire / drops	Sky, no terrain hit within 20 km, terrain/pose failure, axis drift or mechanical limits. It cannot follow a moving vehicle. Mechanical loss switches to STAB and flashes X.
Image rolls / STAB drifts over ground	Roll compensation is intentionally absent. STAB holds a direction; only POINT compensates translation to a ground point.
IR washes out or changes brightness	Try MAN, reset gain/level and brightness/contrast, then adjust. AUTO is scene dependent. Fixed exposure cannot recover clipped detail or guarantee a realistic thermal scene.
No L-code	No acknowledged spot: check laser request, terrain intersection and power. Toggling laser does not guarantee designation of the visible object.
No circle or TTI with MSL	Next missile must be identified as Hellfire. APKWS/unknown hides these cues. TTI also needs a confirmed release and a supported valid range.
Ammunition label differs from rack	It counts the whole identified family across four stations, not one rack. Identification depends on stock station metadata.
Zeroing flashes X	Check ASN power and selected Fly-To waypoint; target direction may be beyond travel. A reachable sky ray instead leaves STAB.

Compatibility and fidelity

The native helpers depend on validated DCS/CockpitBase build signatures and internal interfaces. An update can require a new submod version; do not assume compatibility across DCS builds. This package does not require editing files in the main DCS installation. Multiplayer, VR, other overlays and every seat/render configuration have not been exhaustively validated.

The video is a separate DCS render view and may differ from the main cockpit view in shadows, self-aircraft visibility, rotor depiction or scene detail. There is no separate modeled detector resolution, calibrated temperature, weather attenuation, laser cooling/duty cycle, object tracker, seeker lock, missile guidance API or real impact feedback. Performance varies with the scene, hardware and DCS settings.

Documentation basis: patch 64 runtime scripts, input profiles, native-helper source and automated verification. Patch 64 automated checks passed; its complete live-mission behavior is not certified by those checks. All stated numerical values describe this implementation, not the real aircraft or sensor.